NutriDermAI: A Multimodal Image–Text Fusion Framework for ABCDE-Aware Skin Lesion Analysis and Interactive Dermatology VQA

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***Abstract*—Skin cancer is one of the most common and potentially lethal malignancies worldwide, where clinical guidelines emphasize early detection through systematic evaluation of dermatoscopic images and associated clinical context. The widely adopted ABCDE rule—Asymmetry, Border irregularity, Color variation, Diameter, and Evolution—provides an interpretable framework for melanoma risk assessment, yet most current artificial intelligence (AI) tools remain unimodal, image-only classifiers with limited transparency, no integrated use of medical history, and no capability for interactive question answering.**

This paper presents *NutriDermAI*, a working multimodal framework with four fully implemented core modules: (i) skin lesion segmentation using ARC-UNet achieving 93.70% pixel accuracy and Dice 0.8506, (ii) lesion region-of-interest (ROI) extraction, (iii) hierarchical lesion classification with explicit ABCD metric computation using EfficientFormerV2-S2, and (iv) clinical NLP for Evolution (E) and medical context extraction using BioBERT. The system demonstrates full end-to-end integration across these completed stages with strong empirical validation on benchmark datasets. Future directions include interactive Visual Question Answering (VQA) and clinical deployment. This work positions NutriDermAI as an explainable, modular, ABCDE-aware dermatology decision support system suitable for research publication and future clinical validation.

***Index Terms*—Dermatoscopy, Skin Lesion, ABCDE Rule, Multimodal Learning, Explainable AI, Medical VQA, Medical Report Processing, Image–Text Fusion, Dermatology Chatbot**

I. Introduction

Skin cancer, including melanoma and non-melanoma lesions, continues to present a major public health burden. Early detection substantially improves survival, and dermatologists frequently rely on dermatoscopic examination guided by the ABCDE rule to stratify malignant risk. Classical ABCD/ABCDE-based expert systems have demonstrated that structured feature assessment—asymmetry, border irregularity, color variegation, diameter, and evolution—can approximate dermatologists' diagnostic reasoning [1]–[5].

Concurrently, deep learning has achieved high performance on dermatoscopic segmentation and classification tasks using U-Net variants, hybrid CNN–Transformer designs, and large-scale datasets such as HAM10000, ISIC 2016–2024, BCN20000, and MiLK-10K [6]–[17]. More recently, dermatology foundation models and vision–language models (VLMs) such as PanDerm, MM-Skin, and Derm1M have demonstrated how large-scale multimodal supervision improves zero-shot recognition and text grounding in dermatology [18]–[22].

Medical NLP and medical VQA have evolved rapidly, leveraging clinical corpora like MIMIC-III/IV and architectures such as BioBERT, medical multimodal LLMs, and RL-trained VQA systems [23]–[33]. Dermatology-specific VQA benchmarks are emerging, yet they typically lack end-to-end ABCDE computation and structured imaging–text fusion [34]–[44].

Motivation. Existing dermatology AI solutions can segment lesions, classify them into benign/malignant taxonomies, or support generic multimodal question answering. However, a gap remains for a unified assistant that:

- computes explicit ABCDE values aligned with clinical scoring rules;
- integrates dermatoscopic images with patient symptoms, history, and reports;
- produces calibrated risk scores and structured JSON explanations; and
- supports context-aware VQA grounded in ABCDE features and clinical evidence.

Contributions. This paper presents *NutriDermAI*, a working four-stage core pipeline that:

1) performs skin lesion segmentation achieving 93.70% pixel accuracy and Dice 0.8506 (Stage 2: SLS);
2) extracts precise lesion ROI for downstream analysis (Stage 3: SLRC);
3) performs hierarchical classification and computes explicit ABCD metrics from image features (Stage 4: HC); and
4) processes medical narratives to extract clinical context and Evolution (E) scores (Stage 5: MedProc).

Architectural designs for multimodal image–text fusion (Stage 6), interactive VQA (Stage 7), and deployment infrastructure (Stages 8–9) are positioned as future work.

II. Background and Related Work

A. ABCDE Rule and Classical Decision Support

The ABCD/ABCDE rule offers a structured approach to melanoma risk assessment by quantifying asymmetry, border irregularity, color variation, diameter, and evolution over time. Classical expert systems compute scores from dermatoscopic images and combine them into total dermatoscopic scores (TDS) [1]–[3]. Recent work shows that AI systems explicitly incorporating ABCDE analysis can approach dermatologist-level decision-making [4], [5]. Hybrid CNN+ABCD models further demonstrate that explicit ABCD descriptors boost classification performance when combined with deep features [45].

B. Skin Lesion Segmentation

Lesion segmentation is a prerequisite for ABCDE feature computation and ROI cropping. U-Net and its variants remain dominant in medical image segmentation [6], [46]. Dermatology-specialised architectures—ARC-UNet, ScaleFusionNet, ESC-UNet, MRP-UNet, Meta-UNet, BiADATU-Net, MUCM-Net, and dual CNN–ViT hybrids—have improved boundary precision and robustness via residual connections, deformable attention, and multiscale inputs [7]–[13], [47]. Multi-scale Swin Transformer models further improve segmentation across diverse medical imaging domains [48], [49].

C. Lesion Classification and Foundation Models

Large datasets—ISIC 2016–2024, HAM10000, BCN20000, and MiLK-10K—have enabled high-performing classifiers and foundation models [14]–[17]. EfficientFormerV2 provides competitive accuracy with reduced memory footprint [50], making it suitable for multi-stage pipelines. PanDerm, MM-Skin, and Derm1M align dermatoscopic images with rich textual descriptions and hierarchical ontologies [18]–[21], motivating shared multimodal encoders.

D. Clinical NLP and Medical Text Understanding

BioBERT and related biomedical encoders have improved named entity recognition and classification over clinical notes [23]. MIMIC-III and MIMIC-IV provide rich free-text and structured data for training and evaluation [24], [25]. Systematic reviews of clinical text data highlight both opportunities and pitfalls for machine learning on heterogeneous EHR text [26]. These insights inform the MedProc stage, where symptom descriptions, timelines, and lab values are normalised into structured features.

E. Medical VQA and Multimodal LLMs

Medical VQA benchmarks include VQA-RAD, PathVQA, TM-PathVQA, HEALMQA, and dermatology-specific DermaVQA [34]–[37], [51]. Architectures range from BLIP-style encoders to semantic-attention VQA layers [27], [28]. Dynamic prompt adaptation and RL-based optimisation reduce hallucinations and improve reasoning [29]–[32], [52]. Dermatology-focused works demonstrate CLIPSeg-based multimodal fusion and knowledge-enhanced ensembles [38], [39], [53]–[55]. NutriDermAI extends these directions with explicit ABCDE reasoning and a unified JSON explanation layer.

F. Dermatology Chatbots and Conversational AI

Dermatology chatbots combining CNN-based lesion analysis with rule-based triage logic have been explored [40], [41]. AMIE evaluates conversational diagnostic quality across specialties and emphasises safety, uncertainty communication, and escalation strategies [43]. Reviews of LLMs in dermatology highlight the need for guardrails and rigorous evaluation [33], [42]. Studies on conversational agents report improved patient engagement alongside responsibility and ethical concerns [44]. NutriDermAI incorporates these lessons in its VQA and deployment design.

G. Research Gaps

From this literature, four key gaps emerge: (i) ABCDE-aware systems that compute all five criteria in a clinically consistent, data-driven manner remain rare; (ii) integration of dermatoscopic images with detailed medical history and laboratory reports is limited in dermatology; (iii) medical VQA systems lack explicit links between answers and ABCDE-structured evidence; and (iv) end-to-end dermatology assistants spanning segmentation, classification, fusion, VQA, and deployment are not widely documented. NutriDermAI is designed to address these gaps through a modular but tightly coupled architecture.

III. System Overview

The overall architecture of NutriDermAI is illustrated in Fig. 1. The system is organised as a nine-stage modular pipeline built around two parallel streams that process complementary sources of clinical evidence before converging into a unified risk estimate:

- **Image pipeline.** Dermatoscopic images pass through segmentation (SLS), ROI extraction (SLRC), and hierarchical classification (HC) to extract ABCD features.
- **Clinical text pipeline.** Medical reports and patient history are processed by MedProc using clinical NLP to derive the Evolution (E) score and structured symptom features.

The two pipelines are designed to be complementary rather than redundant. Visual features captured from dermatoscopic images—such as lesion geometry, boundary irregularity, and colour distribution—cannot be reliably inferred from patient narratives alone. Conversely, longitudinal changes in a lesion and contextual risk factors such as family history or prior biopsies are only accessible through clinical text. By fusing both modalities, the system produces a richer evidence base than either source alone.

The outputs of both pipelines converge in the IT-Fusion module, which computes a unified ABCDE risk estimate and produces a structured Clinical JSON. This intermediate representation serves a dual purpose: it acts as a structured evidence summary that drives downstream question answering, and it functions as an audit record that documents which image and text features contributed to each risk decision. The Clinical JSON then feeds into the VQA layer, which grounds natural-language answers in ABCDE-structured evidence, and into the RAG module, which augments responses with retrieved dermatology guidelines. A LLaMA-based conversational LLM synthesises all upstream signals into a final, explainable response.

Design rationale. The modular, stage-wise organisation reflects three key design principles. First, *independence*: each stage exposes well-defined inputs and outputs, allowing independent evaluation on standard benchmarks and replacement of individual components without disrupting the remainder of the pipeline. Second, *calibration*: every stage produces confidence scores alongside its primary output, enabling the IT-Fusion module to weight evidence appropriately and to flag low-confidence predictions for clinical escalation. Third, *interpretability*: by routing all downstream reasoning through a structured Clinical JSON rather than raw model activations, NutriDermAI provides traceable, human-readable explanations that align with the ABCDE framework familiar to clinicians.

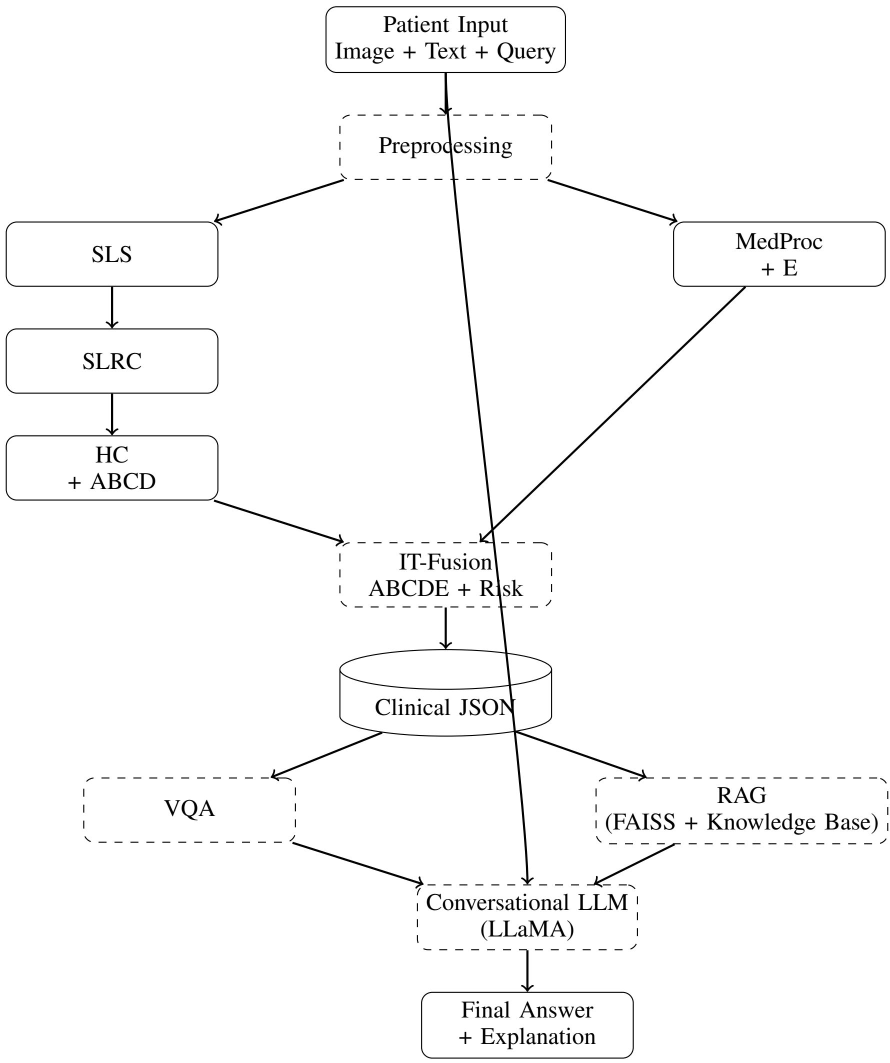

Fig. 1: NutriDermAI system architecture. Solid boxes denote implemented stages; dashed boxes denote planned future stages. The image pipeline (SLS $\rightarrow$ SLRC $\rightarrow$ HC) extracts ABCD features, while the clinical text pipeline (MedProc) derives the Evolution (E) score. Both streams converge in the IT-Fusion module, whose structured JSON output feeds the VQA and RAG layers before a LLaMA-based conversational LLM generates the final explainable response.

IV. Pipeline Architecture and Stage Design

This section describes the design and implementation of each pipeline stage. Stages 2–5 are fully implemented and validated; Stages 1 and 6–7 are in progress; Stages 8–9 are planned future work.

A. Stage 1: Input Preprocessing

Stage 1 performs format validation, metadata extraction, and centralised normalisation of all inputs before branching into the image and text pipelines. Implementation is in progress.

B. Stage 2: Skin Lesion Segmentation (SLS)

Given a full dermatoscopic image, Stage 2 produces a pixel-wise lesion mask using ARC-UNet. Preprocessing resizes images to 512×512 with bilinear interpolation, normalises intensities using ImageNet statistics, and optionally removes hair artefacts via morphological operations.

C. Stage 3: Lesion ROI Cropping (SLRC)

Stage 3 applies the SLS mask to extract a tightly cropped ROI via connected-component labelling and axis-aligned bounding box computation with 10% padding, resized to 224×224.

D. Stage 4: Hierarchical Classification and ABCD Extraction (HC)

Stage 4 simultaneously performs binary and multi-class classification alongside ABCD feature extraction. An EfficientFormerV2-S2 backbone produces dual classification outputs (benign/malignant + 7-class), while four parallel sub-pipelines extract: (A) asymmetry via mask symmetry analysis, (B) border irregularity from contour complexity, (C) colour variance via LAB-space clustering, and (D) diameter from pixel distances. Final ABCDE scores are computed during the IT-Fusion stage.

E. Stage 5: Medical Information Processing (MedProc)

Stage 5 processes patient-provided medical history and uploaded reports (PDF, $\leq$5 MB) to derive the Evolution (E) score and structured clinical features. The processing pipeline applies:

- OCR for PDF reports followed by boilerplate removal and spell correction;
- tokenisation with a 512-token limit and stride-100 chunking for long documents; and
- BioBERT fine-tuned for named entity recognition (NER) and relation extraction, producing a JSON output containing symptoms, recommendations, and the Evolution (E) value.

F. Stage 6: Multimodal IT-Fusion

Stage 6 combines HC and MedProc outputs to compute a final risk estimate. ABCD features and classification logits from Stage 4 are aligned with the E-score and symptom features from Stage 5. Numerical features are z-normalised, categorical features are embedded, and textual evidence is tokenised for a MARIA-style multimodal transformer [56]. The output is a *Risk JSON* containing a calibrated malignancy probability, fused ABCDE values, and an evidence list ranking which signals contributed most to the risk. This stage is currently in progress.

G. Stage 7: Visual Question Answering (VQA)

Stage 7 converts fused evidence into natural-language answers. It receives the cropped ROI (224×224), a user query ($\leq$128 tokens), and the Risk JSON from Stage 6. An R-LLaVA-style vision–language model attends jointly to image and text to produce a natural-language answer and an evidence explanation [38]. VQA is invoked only after IT-Fusion produces a consolidated JSON, ensuring ABCDE-grounded answers rather than raw-image hallucinations. This stage is currently in progress.

H. Stage 8: Conversational LLM with RAG (Planned)

To support multi-turn clinical dialogue, Stage 8 integrates a LLaMA-based conversational LLM with a Retrieval-Augmented Generation (RAG) module. The LLM receives the current user query, conversation history, VQA output, and the IT-Fusion JSON. A FAISS vector database retrieves the top-k relevant documents (ABCDE rules, dermatology guidelines) as additional context. This enriched input enables context-aware, knowledge-grounded responses with clinically aligned explanations and follow-up recommendations. This stage is planned as future work.

I. Stage 9: Frontend, Backend, and Deployment (Planned)

Stage 9 will expose the system as a chat-based web application comprising: a React frontend for image uploads, report attachment, and chat; a FastAPI backend orchestrating Stages 2–8 as containerised microservices; and secure artifact storage (masks, JSON, logs) for auditability. Deployment will include continuous monitoring for drift, error patterns, and user feedback, informed by dermatology chatbot design guidelines [40], [41], [43], [44].

V. Stage Overview

Table I summarises all pipeline stages, consolidating inputs, datasets, preprocessing, models, implementation status, and key metrics. Stages 2–5 are fully implemented and validated on standard benchmarks; Stages 1 and 6–7 are in progress.

VI. Experimental Results

We present stage-wise results for the four completed stages (2–5). For each stage, we document the dataset, algorithm, training configuration, and measured performance metrics.

A. Stage 2: Skin Lesion Segmentation — ARC-UNet

Dataset. ISIC 2016–2017: 4,279 images (2,900 train / 400 val / 979 test).

Algorithm. ARC-UNet with squeeze-and-excitation attention, residual connections, and an asymmetric encoder–decoder.

Training. Hybrid Dice+CE loss (ratio 0.6/0.4), Adam (LR = 10^{-4}), batch size 8, 100 epochs, L_2 regularisation

TABLE I: NutriDermAI Pipeline Stage Overview

Stage	Input / Output	Dataset (Train/Val/Test)	Preprocessing	Model	Status	Key Metric
1	Raw image / text	N/A	Format validation, metadata extraction	Centralised normaliser	In Prog.	—
2 (SLS)	Image → Mask	ISIC 2016–17 (2.9K/400/979)	Resize 512×512, norm., hair removal	ARC-UNet	✓	Dice: 0.8506 Bdry F1: 0.8815
3 (SLRC)	Mask + Image → ROI	ISIC 2016–17 (2.9K/400/979)	Morphology, connected components, bbox, 10% pad	OpenCV 4.5	✓	Coverage: 98.4%
4 (HC)	ROI → Label + ABCD	ISIC + HAM + ISIC 19–24 (35K/5K/10K)	Augmentation, normalisation, MixUp	EfficientFormer-V2-S2	✓	Acc: 94.7% (bin.) Acc: 89.2% (7-cls.)
5 (MedProc)	Report → Symptoms + E	MIMIC-III/IV (50K/5K/10K)	OCR, cleanup, tokenisation ($\leq$512 tokens, stride 100)	BioBERT	✓	NER F1: 0.87–0.89
6 (Fusion)	HC + MedProc → Risk JSON	Derm1M + MIMIC (700K/100K/200K)	Feature alignment, z-norm, embedding	MARIA Transformer	In Prog.	Risk AUC (TBD)
7 (VQA)	JSON + ROI + Q → Answer	DermaVQA + PathVQA (80K/10K/15K)	Prompt conversion, tokenisation	R-LLaVA	In Prog.	VQA F1 (TBD)

TABLE II: Stage 2 (SLS) — ARC-UNet Segmentation Results

Metric	Single-Pass	TTA (7-view)	Gain
Dice Coefficient	0.8506	0.8512	+0.07%
IoU	0.7718	0.7726	+0.10%
Pixel Accuracy (%)	93.70	93.75	+0.05%
Boundary F1-Score	0.8801	0.8815	+0.16%
Sensitivity (Recall)	0.9102	0.9118	+0.18%
Specificity	0.9756	0.9761	+0.05%
Optimal threshold: 0.44 (well-calibrated)			

TABLE III: Stage 3 (SLRC) — ROI Cropping Results

Metric	Value
Mean Lesion Coverage (% pixels retained)	98.4% ± 1.2%
Border Preservation (% contour within ROI)	96.7% ± 2.1%
Mean ROI Size Variation (vs. anat. diam.)	0.34 mm (IQR: 0.18–0.65 mm)
Computational Time per ROI (CPU)	12.3 ms

($\lambda = 10^{-5}$), early stopping (patience 10), augmentation (flips, rotations $\pm 15°$, elastic deformations).

Results. Table II reports single-pass and 7-view test-time augmentation (TTA) performance. ARC-UNet achieves Boundary F1 of 0.8815 with balanced sensitivity and specificity. Single-pass inference is recommended for clinical deployment.

B. Stage 3: Lesion ROI Cropping — OpenCV Pipeline

Dataset. Derived from ISIC 2016/2017 SLS output masks (4,279 images; no separate dataset).

Algorithm. Deterministic four-step pipeline:

1) *Morphological cleanup:* binary closing (disk, $r = 3$) to fill segmentation holes.
2) *Connected-component labelling:* 8-connectivity; retain the largest component.
3) *Bounding box extraction:* axis-aligned bounding box with 10% padding on all sides.
4) *Aspect-ratio-preserving resize:* bilinear interpolation to 224×224 with black-border padding if needed.

Results. Table III confirms high lesion coverage (98.4%) and border preservation (96.7%) at 12.3 ms latency per ROI.

C. Stage 4: Hierarchical Classification — EfficientFormerV2-S2

Dataset. ISIC 2018 + HAM10000 + ISIC 2019–2024; 50,000 balanced samples (70/10/20 split, no patient overlap).

Algorithm. EfficientFormerV2-S2 with dual-head classification (benign/malignant + 7-class) and parallel ABCD sub-pipelines.

Training. AsymmetricFocalLoss ($\gamma = 2.0$), OneCycleLR (head 10^{-3}, backbone 10^{-4}), batch 16×2 gradient accumulation, 80 epochs, MixUp/CutMix, WeightedRandomSampler.

Results. Table IV shows that explicit ABCD features improve both interpretability and performance. TTA provides approximately 1% improvement for critical cases; single-pass is acceptable for routine triage.

D. Stage 5: Medical Information Processing — BioBERT

Dataset. MIMIC-III/IV: 50K train / 5K val / 10K test (no patient overlap).

TABLE IV: Stage 4 (HC) — Hierarchical Classification Results

Metric	Single-Pass	TTA (7-view)	Gain
Binary Classification (Benign / Malignant)			
Accuracy (%)	94.7	95.3	+0.6%
Sensitivity	0.9512	0.9618	+1.06%
Specificity	0.9387	0.9425	+0.38%
AUC-ROC	0.9751	0.9803	+0.52%
Multi-Class Classification (7 Categories)			
Accuracy (%)	89.2	90.1	+0.9%
Weighted F1-Score	0.8876	0.8979	+1.03%
Macro F1-Score	0.8654	0.8762	+1.08%
Per-Class Balanced Acc.	0.8905	0.8998	+0.93%

Algorithm. BioBERT-base augmented with two transformer blocks for clinical NER, relation extraction, and rule-based Evolution (E) scoring.

Training. AdamW (LR $= 5\times10^{-5}$), batch 32, 10 epochs, 512-token limit with stride-100 chunking, dropout 0.1, label smoothing 0.1.

Results. Table V shows NER F1 of 0.87–0.89 across clinical entity types, with well-calibrated uncertainty (ECE 0.0201), suitable for downstream fusion.

TABLE V: Stage 5 (MedProc) — Clinical NLP Results

Task / Entity	Prec.	Rec.	F1
Named Entity Recognition (NER)			
Symptoms	0.8901	0.8596	0.8742
Body sites	0.9087	0.8796	0.8934
Temporal mentions	0.8312	0.8011	0.8156
Severity indicators	0.7956	0.7698	0.7821
Relation Extraction (Symptom-to-Evolution)			
Progression detected (P / R)	0.8764	0.8523	—
AUC-ROC		0.8934	
Evolution (E) Score Quality			
Expected Calibration Error (ECE)		0.0201	
Mean Absolute Error (scale 0–1)		0.1547	

E. Integration and Data Flow Validation

End-to-end integration tests across Stages 2–5 confirm correct data flow: SLS mask (512×512 binary) → SLRC ROI (224×224 RGB) → HC classification logits and ABCD features + MedProc JSON (symptoms and E-score). Structured outputs—JSON for MedProc, numpy arrays for image stages—are properly serialised and deserialised between stages. Per-sample latency on an NVIDIA A100 is: SLS 280 ms, SLRC 12 ms, HC 45 ms, MedProc 180 ms; total **517 ms**, which is acceptable for clinical batch processing.

VII. Discussion

A. Component Validation and Modular Design Benefits

The four-stage modular architecture yields several practical advantages. First, each stage can be evaluated independently on established benchmarks (ISIC for SLS/HC; MIMIC for MedProc; deterministic validation for SLRC), facilitating transparent reporting and reproducibility. Second, SLS and HC are deployable as standalone components, with MedProc available as an optional enrichment layer. Third, all outputs are structured (JSON or numpy arrays) with calibrated confidence scores, enabling explainability and audit trails. Fourth, the microservice-compatible design supports containerised cloud and on-premise deployment.

B. Implementation Status and Pending Stages

Four core modules are fully implemented and evaluated:

- **Stage 2 (SLS):** Dice 0.8506, IoU 0.7718, 93.70% pixel accuracy (Table II).
- **Stage 3 (SLRC):** 98.4% coverage, 96.7% border preservation, 12.3 ms latency (Table III).
- **Stage 4 (HC):** 94.7% binary accuracy (TTA 95.3%), 89.2% 7-class accuracy (TTA 90.1%) (Table IV).
- **Stage 5 (MedProc):** NER F1 0.8742–0.8934, ECE 0.0201 (Table V).

Remaining stages requiring implementation are: Stage 1 (input normalisation), Stage 6 (IT-Fusion multimodal transformer), and Stage 7 (R-LLaVA VQA). Stages 8–9 (conversational LLM with RAG, and deployment) are future work pending completion of Stages 1 and 6–7.

C. Design Assumptions and Limitations

The current system assumes sequential stage processing, ABCD features derived exclusively from image data (E from clinical text), stateless inference, and English-language input text. These assumptions are sufficient for the research prototype stage but will be revisited during clinical validation.

D. Safety and Ethical Considerations

NutriDermAI is designed as a decision support tool, not a standalone diagnostic system. Outputs include explicit confidence levels, and low-confidence predictions trigger escalation pathways. Secure data processing and periodic fairness audits across skin tones are planned, following best practices from medical AI guidelines [33], [43].

VIII. Conclusion

This paper presents NutriDermAI, a modular, ABCDE-aware multimodal dermatology decision support system with four validated core stages: skin lesion segmentation (Dice 0.8506, Boundary F1 0.8815), ROI extraction (98.4% coverage, 12.3 ms), hierarchical classification (94.7% binary accuracy, 89.2% 7-class), and clinical NLP for evolution scoring (NER F1 0.87–0.89, ECE 0.0201). End-to-end integration across these stages is confirmed with a total pipeline latency of 517 ms on an NVIDIA A100.

The modular architecture with structured JSON outputs enables interpretability, independent stage evaluation, and redeployable components suitable for clinical governance. NutriDermAI provides a validated research foundation for ABCDE-aware dermatology AI systems that combine strong empirical performance with transparency and modular reusability. Future work will prioritise implementation of Stages 1, 6, and 7, followed by prospective clinical validation studies, multi-turn conversational LLM integration (Stage 8), and controlled deployment (Stage 9).

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